

TEMPLATE

KEY PROJECT INFORMATION & PROJECT DESIGN DOCUMENT (PDD)

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VERSION **v. 1.2**

RELATED SUPPORT

- TEMPLATE GUIDE Key Project Information & Project Design Document v.1.2

This document contains the following Sections

Key Project Information

0 –_Description of project

0 – Application of approved Gold Standard Methodology (ies) and/or demonstration of SDG Contributions

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Appendix 1 – Safeguarding Principles Assessment (mandatory)

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KEY PROJECT INFORMATION

GS ID of Project	GS11351
Title of Project	Thuan Nhlen Phong Wind Farm
Time of First Submission Date	14/09/2021
Date of Design Certification	14/06/2022
Version number of the PDD	5.0
Completion date of version	30/10/2022
Project Developer	Thuan Nhlen Phong 1 Wind Power Company Limited
Project Representative	Thuan Nhlen Phong 1 Wind Power Company Limited
Project Participants and any communities involved	Thuan Nhlen Phong 1 Wind Power Company Limited
Host Country (ies)	Viet Nam
Activity Requirements applied	☐ Community Services Activities ☒ Renewable Energy Activities ☐ Land Use and Forestry Activities/Risks & Capacities ☐ N/A
Scale of the project activity	☐ Micro scale ☐ Small Scale ☒ Large Scale
Other Requirements applied	N/A
Methodology (ies) applied and version number	ACM0002 - Grid-connected electricity generation from renewable sources (version 20.0)
Product Requirements applied	☒ GHG Emissions Reduction & Sequestration ☐ Renewable Energy Label ☐ N/A
Project Cycle:	☐ Regular ☒ Retroactive

Table 1 – Estimated Sustainable Development Contributions

Sustainable Development Goals Targeted	SDG Impact (defined in B.6.)	Estimated Annual Average	Units or Products
13 – Climate Action	Emission Reduction	85,736	tCO ₂ /year or GS VERs
8 – Decent Work and Economic Growth	Full and productive employment and decent work for all (All employees have access to training, healthcare, insurances and better income)	11	employees
7 – Affordable and Clean Energy	Ensure access to affordable, reliable, sustainable and modern energy for all (MWh of renewable energy generated)	95,661	MWh

SECTION A. DESCRIPTION OF PROJECT

A.1 Purpose and general description of project

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Thuan Nhien Phong Wind Farm Project (hereafter “Proposed Project” or “Project Activity”) is a greenfield grid-connected wind power plant which is located at Hoa Thang commune, Bac Binh district, Binh Thuan province, Vietnam.

The proposed project is owned by Thuan Nhien Phong 1 Wind Power Company Limited (hereafter “TNP”). The project activity has a total installed capacity of 30.4 MW, with a predicted power generation of 95,661 MWh per annum. The project activity involves the installation of 8 Wind Turbine Generators (WTGs), with each has an installed capacity of 3.8 MW.

Prior to the implementation of the project activity, electricity in Vietnam is generated mainly from fossil fuel sources and is solely distributed to consumers by Vietnam Electricity (EVN) via the unique national electricity grid. The baseline scenario of the project activity is the same as the scenario existing prior to the start of implementation of the project activity.

The entire electricity generated by the project activity will be fed into the Vietnam national grid (hereafter referred as “the grid”) without any Greenhouse Gas (GHG)

emissions. The operational lifetime of the wind power project is 25 years¹. The electricity generation will result 85,736 tCO₂e/year and total emission reduction will be 428,680 tCO₂e for the first crediting period of 5 years.

The project activity contributes to the substantial development as follows:

Economic well-being

Vietnam has been continuously facing power shortage due to increasing demand and inadequate supply. Hence, the government has laid out plans to nearly double the country's installed generation capacity in its base scenario, to cope with the expected surge in power demand. The project will contribute towards country's power demand by supplying electricity directly to the national grid. Furthermore,

- The project will reduce the dependency of Vietnam on coal, fossil fuel and electricity importation as well as negative impact on the foreign exchange. It will contribute to strengthen the renewable energy sector, especially wind energy sector in Vietnam.
- The project gives a boost to the wind energy sector and push R&D in this field thus improving national economy through innovations.
- The project contributes to poverty alleviation through employment opportunities during project construction & operation as well as towards nation's tax revenues

Social well-being

- The project has provided / will provide job opportunities to local people during implementation, commissioning and operation of the wind project.
- Frequent visits by engineers, professionals and industrialists to the project sites during project execution will create positive impacts on the economy of the communes and improve residents' lifestyle.
- The wind project supports the "revised Power Development Plan VIII (PDP VIII)" in increasing the electricity generation by using renewable energy sources. Also, development of infrastructure like road network and street lightings etc.

¹ As per "Option (c)" of "TOOL10: Tool to determine the remaining lifetime of the equipment (Version 01)"

Environmental well-being

- Wind energy is one of the cleanest renewable sources and does not involve any fossil fuel. Thus, the wind project contributes to environmental well-being without causing any negative impact on health and surrounding environment.
- The wind project does not contribute to the carbon emissions, harmful pollutants and suspended airborne particulate matter associated with coal and fossil fuel fired power plants.

In conclusion the project activity will contribute positively towards sustainable development of Viet Nam.

The chronology of project implementation as follows:

Date	Significant Actions
30/10/2020	Date of EPC (Engineering, Procurement & Construction) contract signed – start date of the project
23/11/2020	Authority's approval on the basic design
24/11/2020	SCADA Agreement
10/12/2020	Power Purchase Agreement (PPA) name transfer
28/01/2021	Measurement & Verification Agreement with EVN
10/09/2021	Submission to GS for project listing
29/10/2021	Expected commercial operation

All the supporting documents for the above key actions are submitted together with this consultation report.

A.1.1. Eligibility of the project under Gold Standard

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The project activity complies with all the eligibility criteria as per section 3.1.1 of GS4GG Principles & Requirements, version 1.2 as follows:

(a) Types of Project	
As per section 4. Eligibility Principles and Requirements of GS4GG Principles & Requirements, version 1.2	
Eligibility Criteria	Compliance by Project Activity
A Project type is automatically eligible for Gold Standard Certification if there are approved Gold Standard Activity	The project type is a large-scale wind project which is an eligible project type as per section 2. Eligible Project Types

Requirements and/or Gold Standard Impact Quantification Methodologies associated with it or as referenced in Gold Standard Product Requirements. These are published to the Gold Standard website and shall be followed where provided for a given Project type	under Renewable Energy Activity Requirements (Version 1.3). The project activity applies CDM methodology ACM0002 - Grid-connected electricity generation from renewable sources (Version 20.0), which is an approved methodology under Gold Standard.
For Project types that are not automatically eligible, a Project Developer may submit to Gold Standard for approval. This shall be done at minimum as part of the Preliminary Review, though it is recommended to engage with Gold Standard earlier to establish the criteria and requirements for approval	The project is automatically eligible as per "GS Activity Requirements".
Project types applying for Gold Standard approval are referred to the Gold Standard Vision and Mission. The Project Developer shall demonstrate how the Project would contribute to these and how the Gold Standard for the Global Goals Requirements would be met in their application for approval	Wind project is an approved project type as per "GS Activity Requirements" and does not require approval from Gold Standard.
In reviewing a new Project type for approval, Gold Standard may establish new Requirements to be met by the Project in order to achieve Gold Standard Design Certification and ongoing Gold Standard Certified Project status. Where required, Gold Standard shall engage expert peer reviewers to establish these	Wind project is an approved project type as per "GS Activity Requirements" and does not require approval from Gold Standard.

Requirements, at the Project Developer's expense	
Gold Standard does not support Project types associated with geo-engineering or energy generated from fossil fuel or nuclear, fossil fuel switch, or any project that supports, enhances or prolongs such energy generation. In certain cases, concerning energy efficiency involving fossil fuels (for example, LPG stoves), an exception is made. This is captured in the relevant Activity Requirements, Gold Standard Approved Methodologies and/or Product Requirements	This project activity is not associated with geo-engineering or energy generated from fossil fuel or nuclear, fossil fuel switch. Also, the project activity does not support, enhance or prolong such energy generation.
As per section 2. Eligible Project Types & 3. General Eligibility Criteria of "Renewable Energy Activity Requirements", version 1.4	
New Gold Standard Verified Emission Reductions (GS VER) or Gold Standard labels for Certified Emission Reductions (GS CER), Renewable Energy projects connected to national or a regional electricity grid must be located in a; a. Least Developed Country (LDC), Small Island Developing State (SIDS) or a Land Locked Developing Country (LLDC). b. Low Income and Low Middle-income country where the penetration level of the proposed Renewable Energy	a. Not applicable b. The proposed project is located at Viet Nam which is categorized as Low Middle-Income Country as per the "The World Bank" ² data. In early Aug 2021, the penetration level of wind energy technology is 1.79% (i.e.819 MW ³) of the total installed capacity (i.e.48,573 MW ⁴) of Viet Nam. This is less than 5% as per the requirement.

² <https://datahelpdesk.worldbank.org/knowledgebase/articles/906519>

³ <https://moit.gov.vn/phat-trien-ben-vung/21-nha-may-dien-gio-voi-tong-cong-suat-la-819-mw-da-vao-van-.html>

⁴ EVN Annual Report 2018 (<https://en.evn.com.vn/d6/news/-Annual-Report-2018-6-13-2118.aspx>)

Technology type is less than 5% of the total grid installed capacity, at the time of the first submission to preliminary review	
Certain Impact Quantification methodologies allow projects to account for a Suppressed Demand scenario when establishing a baseline. In such cases, the application of the Suppressed Demand baseline is limited to small scale and microscale projects.	The proposed project doesn't use Suppressed Demand Baseline.
A single Renewable Energy project may potentially pursue any number and combination of Certified Impact Statements or Products. However, certain Product Requirements, which supersede the generic requirements stated in this document can set restrictions on co-issuance of Certified Impact statements or Products. For instance, GS VERs or GS CERs with REC labels cannot be claimed for the same MWh	The proposed project will claim only GS-VER for the MWh supplied to the national grid.
(b) Location of Project	
The project activity is located at Viet Nam	
(c) Project Area, Project Boundary and Scale	
The project area is 16.4 hectares. As per chosen CDM methodology ACM0002 (version 20.0), The spatial extent of the project boundary includes the project activity and all power plants connected physically to the electricity system that the proposed project is also connected. The detailed on the project boundary is presented in Section B.3 of this PDD. The installed capacity of the project activity is 30.4 MW so it's a large-scale project.	
(d) Host Country Requirements	

The proposed project activity meets all the legal, environmental, ecological and social regulations through obtaining various approvals & license from authorities such as EIA approval before implementation, Land use, construction & grid connection approvals, Investment related license and so on. Also, the project activity contributes to the national and local level sustainable development as defined under section A.1.
(e) Contact Details The contact details of the Project Participants are included under Appendix 2 of this GS PDD.
(f) Legal Ownership Thuan Nhlen Phong 1 Wind Power Company Limited is the legal owner who has full legal right to control and operate the project activity. All the legal ownership documents such as business registration, investment license, PPA, etc. will be uploaded in sustain-cert for design review.
(g) Other Rights None.
(h) Official Development Assistance (ODA) Declaration No public funds or ODA is involved in this project activity. An ODA declaration has been signed by the PP and submitted to GS.

Eligibility criteria defined under PDD guide	
Eligibility Criteria	Compliance by Project Activity
Confirm that the project is not registered with any other voluntary or compliance schemes.	The proposed project activity is not registered with any other voluntary or compliance schemes.
Demonstrate the activity is NOT located in a host country, region, locality or state that has an emission reduction cap enforced OR has the possibility to trade	The host country of proposed project activity is Vietnam which is a non-annex country ⁵ . Hence, no emission reduction cap enforced as well as no emission

⁵ <https://unfccc.int/process/parties-non-party-stakeholders/parties-convention-and-observer-states>

emissions that include the scope of the proposed project	trading system implemented in the host country.
Demonstrate that no potential for double counting of impacts if the Project Area overlaps with that of another Gold Standard or other voluntary or compliance standard programme of a similar nature.	Since project owner keeps confidential and secured all the project related documents, the same nature of project will be unlikely pursue with any other voluntary or compliance standard programme. Also, project owner will continuously monitor through searching in different registries for similar scope of project registered from same location.
Demonstrate that the project is in compliance with applicable Host Country's legal, environmental, ecological and social regulations	The proposed project activity is obtaining all the authorities' approvals to comply with legal, environmental, ecological and social regulations before beginning the implementation.

A.1.2. Legal ownership of products generated by the project and legal rights to alter use of resources required to service the project

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Thuan Nhien Phong 1 Wind Power Company Limited is the legal owner of the proposed project and has legal rights concerning changes in the use of resources required to service the project.

A.2 Location of project

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The proposed project is located in Hoa Thang commune, Bac Binh district, Binh Thuan province, Vietnam.

The geo-coordinates of WTGs are as follow:

WTGs	Latitude	Longitude
1	11° 2'55.10"	108°23'55.53"
2	11° 3'4.01"	108°24'15.53"
3	11° 3'6.81"	108°24'42.13"
4	11° 3'8.11"	108°24'57.83"

WTGs	Latitude	Longitude
5	11° 3'10.71"	108°25'10.73"
6	11° 3'22.71"	108°24'51.03"
7	11° 3'30.41"	108°24'37.03"
8	11° 3'38.51"	108°24'24.93"



A.3 Technologies and/or measures

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The proposed project activity involves the implementation of 30.4 MW grid-connected wind power plant with 8 nos. of 3.8MW GE-137 to convert wind energy into rotating energy of the blades and converting that rotating energy into electrical energy by the generator, which will be supplied to the national grid through the transmission line.

The electricity supplied to the grid will be monitored through the energy meter installed at switching station. They are digital meters with an accuracy level of Class 0.2 to measure the export and import of electricity of Thuan Nhien Phong Wind Farm. The technical parameters of the major equipment as follows:

Parameters	Value	Units
Turbines model	GE-137	–
Rated power output	3.8	MW
Number of WTGs	8	–
Hub height	131.4	m
Rotor		
Diameter	137 m	m
Swept area	675	m ²
Rotational direction	Clockwise (front view)	–

A.4 Scale of the project

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As per section 9.1.2 of “GHG Emission Reduction & Sequestration Requirements”, all projects exceeding the small-scale thresholds are defined as large scale. The small-scale threshold is 15 MW(e) for the renewable energy projects in accordance with “CDM project standard for project activities”.

The total installed capacity of the project activity is 30.4 MW which is greater than 15 MW. Hence, the proposed project is Large Scale.

A.5 Funding sources of project

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Based on the design approval by Industrial & Commercial Facilities division of Binh Thuan’s Peoples Committee for the total investment and the loan facility agreement for debt loan, the funding sources for the total investment are as follows:

Equity – 37%

Debt Loan – 63%

The project doesn’t receive any ODA funding.

SECTION B. APPLICATION OF APPROVED GOLD STANDARD METHODOLOGY (IES) AND/OR DEMONSTRATION OF SDG CONTRIBUTIONS

B.1. Reference of approved methodology (ies)

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Applied Methodology
ACM0002: Grid-connected electricity generation from renewable sources (Version 20.0)
Related Tools
Tool01 – Tool for the demonstration and assessment of additionality (version 07.0.0)
Tool07 – Tool to calculate the emission factor for an electricity system (version 07.0)
Tool10 – Tool to determine the remaining lifetime of equipment (version 01)

B.2. Applicability of methodology (ies)

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Eligibility Criteria as per ACM0002	Compliance by Project Activity
3. This methodology is applicable to grid-connected renewable energy power generation project activities that: (a) Install a Greenfield power plant; (b) Involve a capacity addition to (an) existing plant(s); (c) Involve a retrofit of (an) existing operating plants/units; (d) Involve a rehabilitation of (an) existing plant(s)/unit(s); or (e) Involve a replacement of (an) existing plant(s)/unit(s).	The proposed project activity is a greenfield grid connected renewable (wind) energy power plant.
4. The methodology is applicable under the following conditions:	The proposed project activity is a new installation of wind power plant. It doesn't involve any

(a) The project activity may include renewable energy power plant/unit of one of the following types: hydro power plant/unit with or without reservoir, wind power plant/unit, geothermal power plant/unit, solar power plant/unit, wave power plant/unit or tidal power plant/unit; (b) In the case of capacity additions, retrofits, rehabilitations or replacements (except for wind, solar, wave or tidal power capacity addition projects) the existing plant/unit started commercial operation prior to the start of a minimum historical reference period of five years, used for the calculation of baseline emissions and defined in the baseline emission section, and no capacity expansion, retrofit, or rehabilitation of the plant/unit has been undertaken between the start of this minimum historical reference period and the implementation of the project activity.	capacity additions/ retrofits/ rehabilitations/ replacements the existing plant.
5. In case of hydro power plants, one of the following conditions shall apply:	Not applicable. Proposed project acidity is a wind power plant.
6. In the case of integrated hydro power projects, project proponent shall:	Not applicable. Proposed project acidity is a wind power plant.
7. The methodology is not applicable to: (a) Project activities that involve switching from fossil fuels to renewable energy sources at the site of the project activity, since in this case the baseline	Not applicable. The proposed project activity neither involve switching from fossil fuels to renewable energy sources nor biomass fired power plant.

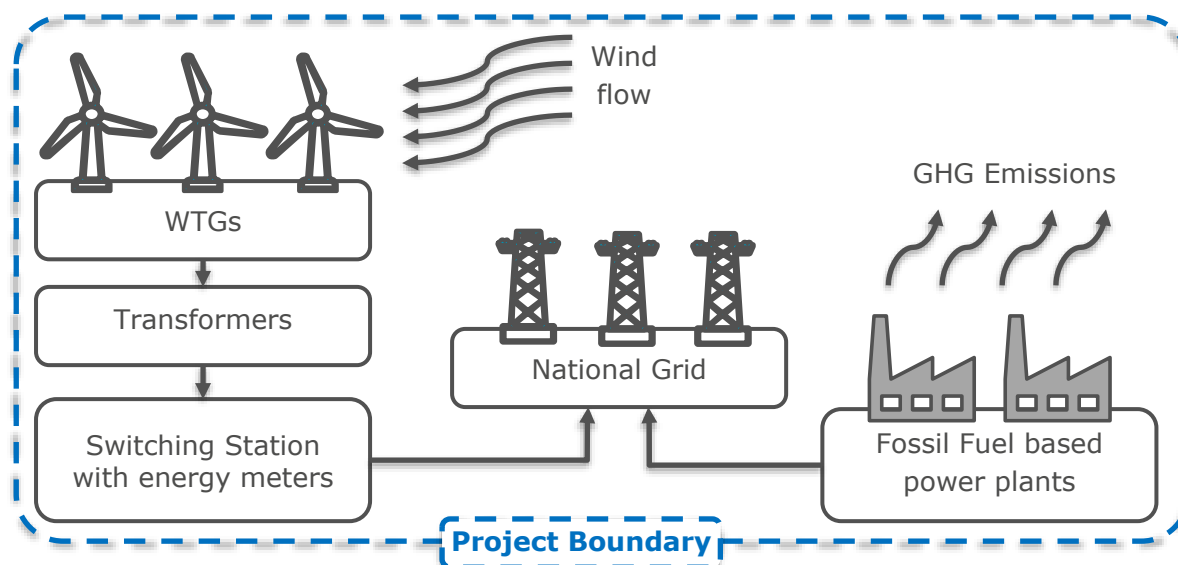
may be the continued use of fossil fuels at the site; (b) Biomass fired power plants/units.	
8. In the case of retrofits, rehabilitations, replacements, or capacity additions, this methodology is only applicable if the most plausible baseline scenario, as a result of the identification of baseline scenario, is "the continuation of the current situation, that is to use the power generation equipment that was already in use prior to the implementation of the project activity and undertaking business as usual maintenance".	The proposed project activity is a new installation of wind power plant. It doesn't involve any capacity additions/ retrofits/ rehabilitations/ replacements the existing plant.

B.3. Project boundary

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As per the chosen methodology ACM0002, the spatial extent of the project boundary includes the Thuan Nhien Phong Wind Farm and all power plants connected physically to the national electricity grid to which the proposed project is also connected.

The flow diagram of the project boundary is shown in Figure below.



Source	GHGs	Included?	Justification/Explanation
Baseline	CO ₂	Yes	Main emission source
	CH ₄	No	Minor emission source

	fossil fuel fired power plants that are displaced due to the project activity	N ₂ O	No	Minor emission source
Project Activity	For dry or flash steam geothermal power plants, emissions of CH ₄ and CO ₂ from non-condensable gases contained in geothermal steam	CO ₂	No	The proposed project activity is a wind power plant and thereby all these sources are not applicable.
		CH ₄	No	
		N ₂ O	No	
	For binary geothermal power plants, fugitive emissions of CH ₄ and CO ₂ from non-condensable gases contained in geothermal steam	CO ₂	No	
		CH ₄	No	
		N ₂ O	No	
	For binary geothermal power plants, fugitive emissions of hydrocarbons such as n-butane and isopentane (working fluid) contained in the heat exchangers	CO ₂	No	
		CH ₄	No	
		N ₂ O	No	
	CO ₂ emissions from combustion of fossil fuels for electricity generation in solar thermal power plants and geothermal power plants	CO ₂	No	
		CH ₄	No	
		N ₂ O	No	
	For hydro power plants, emissions of CH ₄ from the reservoir	CO ₂	No	
		CH ₄	No	
		N ₂ O	No	

B.4. Establishment and description of baseline scenario

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As per chosen methodology ACM0002 (Version 20.0), section 5.2.1, clause 22:

If the project activity is the installation of a Greenfield power plant, the baseline scenario is electricity delivered to the grid by the project activity would have otherwise been generated by the operation of grid-connected power plants and by the addition of new generation sources, as reflected in the combined margin (CM) calculations described in "TOOL07: Tool to calculate the emission factor for an electricity system".

Since the proposed project activity involves the installation of greenfield grid connect wind power plant, the baseline scenario of the project is to provide an equal amount of electricity provided by the national grid where the proposed project is also connected. EVN⁶ is the sole operator of the Viet Nam national electricity grid where all the power plants in Viet Nam are physically connected to and the proposed project activity is not outside of that system.

The combined margin emission factor of the national grid ($EF_{grid,CM,y}$) is calculated as per "Tool to calculate the emission factor for an electricity system (Version 07.0)" and used to calculate the baseline emissions of the proposed project activity. Data to calculate $EF_{grid,CM,y}$ for wind power projects is published (CV 263/BDKH⁷) by Department of Climate Change - Ministry of Natural Resources and Environment on 12/03/2020.

B.5. Demonstration of additionality

As per chosen methodology ACM0002 (Version 20.0), "Tool for the demonstration and assessment of additionality (Version 07.0.0)" shall be used to demonstrate the additionality of the proposed project activity step-by-step as follows:

Step 0: Demonstration whether the proposed project activity is the first-of-its-kind

The proposed project is an onshore wind power project located in Vietnam and it's not a first-of-its-kind.

Step1. Identification of alternatives to the proposed project activity consistent with current laws and regulations

In accordance with the chosen methodology ACM0002 version 20.0, paragraph 22 states:

"if the project activity is the installation of a Greenfield power plant, the baseline scenario is electricity delivered to the grid by the project activity would have otherwise been generated by the operation of grid connected power plant and by the addition of new generation sources."

⁶ <https://en.evn.com.vn/>

⁷ [http://dcc.gov.vn/van-ban-phap-luat/1059/Nghien-cuu,-xay-dung-heso-phat-thai-\(EF\)-cua-luoi-dien-Viet-Nam-\(K%C3%A8m-CV-263/BDKH\).html](http://dcc.gov.vn/van-ban-phap-luat/1059/Nghien-cuu,-xay-dung-heso-phat-thai-(EF)-cua-luoi-dien-Viet-Nam-(K%C3%A8m-CV-263/BDKH).html)

Since the baseline scenario is defined by the chosen methodology, no further analysis is required to carry out to identify the plausible alternatives. However, the identification of plausible alternatives has been considered in order to demonstrate the compliance towards the Tool 01:

Sub-step 1a: Define alternatives to the project activity

In the absence of the proposed project activity, plausible and credible project activities to the proposed project activity are as below:

Alternative 1: The proposed project activity not undertaken as a GS-VER project activity

Alternative 2: Continuation of the current situation (no project activity or other alternatives undertaken)

Sub-step 1b: Consistency with mandatory laws and regulations

The above plausible alternatives do not have any obstacles due to national or local laws and regulations since the proposed project is obtaining all the necessary authorities' approval throughout its implementation and operation, it doesn't face any difficulties due to mandatory laws and regulations. In terms of the second alternatives, it's currently under operation and thereby no effort is required. Hence these alternatives are consistent with mandatory laws and regulations.

Since wind projects are green power without polluting the environment, no concern to operate from pollution control board. Second alternative has been chosen as an appropriate baseline alternative for this project activity in line with the methodology. Hence, both alternatives are found to comply with the mandatory laws and regulations taking into account the enforcement of the legislations in the region or country and EB decisions on national and/or sectoral policies and regulations.

Step 2: Investment Analysis

Determine whether the proposed project activity is not:

- a) The most economically or financially attractive; or
- b) Economically or financially feasible, without the revenue from the sale of emission reductions.

The following steps are used to determine the proposed project activity is not financially attractive or feasible.

Sub-step 2a: Determine appropriate analysis method

As per “Tool for the demonstration and assessment of additionality” (version 07.0.0), paragraph 32 states:

“If the CDM project activity and the alternatives identified in Step 1 generate no financial or economic benefits other than CDM related income, then apply the simple cost analysis (Option I). Otherwise, use the investment comparison analysis (Option II) or the benchmark analysis (Option III).”

Since the project owner couldn’t obtain the financial data regarding the investment of at least one alternative available, Options I & II are not applicable for the proposed project activity. Hence, the most appropriate financial analysis would be Option III (benchmark analysis) where the returns on equity investment from the project activity is compared with nominal benchmark.

Sub-step 2b: Option III. Apply benchmark analysis

The benchmark analysis approach has been chosen to assess the additionality of the proposed project activity.

Identified Financial/Economic Indicator:

The expected return on equity (Equity IRR) was selected as the financial indicator to assess the financial additionality of the proposed project activity as per Methodological tool for Investment analysis (version 10.0).

Selection of Appropriate Benchmark:

Since Equity IRR has been chosen, the default benchmark is adopted from the methodological tool for Investment Analysis, Version 10.0, Appendix.

The default benchmark for Group 1 for Vietnam is 12.6%

Sub-step 2c: Calculation and comparison of financial indicators (only applicable to Options II and III)

As per the Methodological Tool: Investment Analysis, Version 09.0, the general issued in calculation and presentation are discussed as follows:

General issues as per investment analysis tool	Justification
6. The period of assessment should not be limited to the proposed crediting period of the CDM project activity.	The project activity has chosen 25 years (project lifetime) for the IRR calculation.

Both project internal rate of return (IRR) and equity IRR calculations should reflect the period of expected operation of the underlying project activity (technical lifetime) and if a shorter period than the technical lifetime is chosen, the investment analysis shall be conducted for at least 10 years and include the fair value of the project activity assets at the end of the assessment period. The IRR calculation may include the cost of major maintenance and/or rehabilitation if these are expected to be incurred during the period of assessment.	Fair value is considered as zero in accordance with the following explanation: As per Annex 2 of "GUIDING REGULATION ON MANAGEMENT, USE AND DEPRECIATION OF FIXED ASSETS" issued by Ministry of Finance, Straight-line depreciation method defines: $\text{Annual average rate of depreciation for the fixed assets} = \frac{\text{Primary price of fixed assets}}{\text{Time of depreciation}}$ Where, the default value for maximum time of depreciation for wind power assets is 20 years which is mentioned under Annex 1 in the same guideline. Since Thuan Nhien Phong Wind Farm has been chosen 25 years for computation, no fair value is considered.
7. The fair value of any project activity assets at the end of the assessment period shall be included as a cash inflow in the final year. The fair value should be calculated in accordance with local accounting regulations where available, or international best practice. It is expected that such fair value calculations will include both the book value of the asset and the reasonable expectation of the potential profit or loss on the realization of the assets.	Since the IRR calculation has been used the entire project lifetime of 25 years, zero fair value has been considered.
8. The discount rate used in the investment comparison analysis shall be determined following the requirements as set out in this tool	No discount is considered in the IRR computation.

for the calculation of IRR benchmarks in section 6 below.	
9. The weighted average costs of capital (WACC) and the cost of equity provided in the Appendix or calculated using Capital Asset Pricing Model (CAPM) are post-tax IRR benchmarks, and investment analysis shall be conducted with post-tax cash flows. Depreciation, and other non-cash items related to the project activity, which have been deducted in estimating gross profits on which tax is calculated, shall be added back to net profits for the purpose of calculating the financial indicator (e.g. IRR, NPV). The cash flow effects of taxation should be included in the IRR/NPV calculation.	The investment analysis has been conducted with the post-tax cash flows. There are no depreciation and other non-cash items are not considered in the calculation.
10. Input values used in all investment analysis shall be valid and applicable at the time of the investment decision taken by the project participant. The DOE is therefore expected to validate the timing of the investment decision and the consistency and appropriateness of the input values with this timing. The DOE should also validate that the listed input values have been consistently applied in all calculations.	The input values used in all investment analysis has been valid and applicable at the time of the investment decision taken by the project participant. The date of EPC signed on 30 th October 2020 is considered as the investment decision date Therefore, the default benchmark for IRR has been used from Methodological Tool: Investment Analysis, Version 10.0.

11. In the case of project activities for which implementation ceases after the commencement and where implementation is recommenced due to consideration of the CDM the investment analysis should reflect the economic decision-making context at point of the decision to recommence the project. Therefore, capital costs incurred prior to the revised project activity start date can be reflected as the recoverable value of the assets, which are limited to the potential reuse/resale of tangible assets	This is not applicable as the implementation of project activity is not ceased but delayed due to pandemic outbreak.
12. Project participants shall supply spreadsheet versions of all investment analysis. All formulas used in this analysis shall be readable and all relevant cells shall be viewable and unprotected. The spreadsheet will be made available to the Board, UNFCCC secretariat and others contracted to assess the request for registration on behalf of the Board including assigned members of the Registration and Issuance Team. In cases where the project participant does not wish to make such a spreadsheet available to the public an exact read-only or PDF copy shall be provided for general publication. In case the project	The spreadsheet is available during the validation of the proposed project activity.

participant wishes to black-out certain elements of the publicly available version, a clear justification for this shall be provided to the secretariat by the DOE when requesting registration.	
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The values used in investment analysis were valid and applicable at the time of the investment decision taken by the management. The key values used to calculate the return on equity as follows:

Exchange Rate: 1USD = 23,175 VND (Source: State Bank of Vietnam⁸)

Description of Parameters	Values	Unit	Reference
Total installed capacity	30	MW	Investment Policy
Project lifetime	25	years	Methodological Tool 10: Tool to determine the remaining lifetime of equipment
Electricity Generation			
Annual generation (kWh)	115	mil kWh	Energy Yield Assessment (EYA) Report
Electricity Tariff	0.0780	USD/kWh	Power Purchase Agreement (PPA)
O&M Cost			
O&M Annual Base Fee	0.56	mil USD/Year	O&M Contract (Section 7.1)
Project Costing			
Total Project cost	51.35	mil USD	Calculated value
Equity Share	36.97	%	Calculated value
Debt Share	63.03	%	Calculated value
Equity Amount	18.99	mil USD	Capital Investment - Equity Share
Debt (Bank Loan)			
Debt Amount	32.36	mil USD	Loan Facility Agreement
Interest Rate	10.5%	% per year	Notification on Credit Policy
Income Tax			
VAT	10	%	https://moj.gov.vn

⁸ https://www.sbv.gov.vn/TyGia/faces/Aiber.jspx?_afLoop=33777021381882577&_afWindowMode=0&_adf.ctrl-state=49df93rbs_106

CIT	20	%	https://moj.gov.vn
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Outcome of Investment Analysis:

By considering all the above values, the equity IRR for the proposed project activity as follows:

Benchmark (Investment Analysis Tool)	Estimated Equity IRR (without carbon revenue)
12.60	11.10%

The equity IRR is estimated in accordance with multiple documents such as loan agreement, credit policy, O&M contract, PPA, etc. for a period of project activity's technical lifetime (25 years). From the above table, the estimated equity IRR is below than the Benchmark and thereby justifying that the investments are not financially attractive. Therefore, the project activity is an additional and not a Business-As-Usual (BAU).

Sub-step 2d: Sensitivity Analysis

As per the Methodological Tool: Investment Analysis, Version 10.0, paragraph 27 states:

"Variables, including the initial investment cost, that constitute more than 20% of either total project costs or total project revenues should be subjected to reasonable variation (all parameters varied need not necessarily be subjected to both negative and positive variations of the same magnitude), and the results of this variation should be presented in the PDD and be reproducible in the associated spreadsheets."

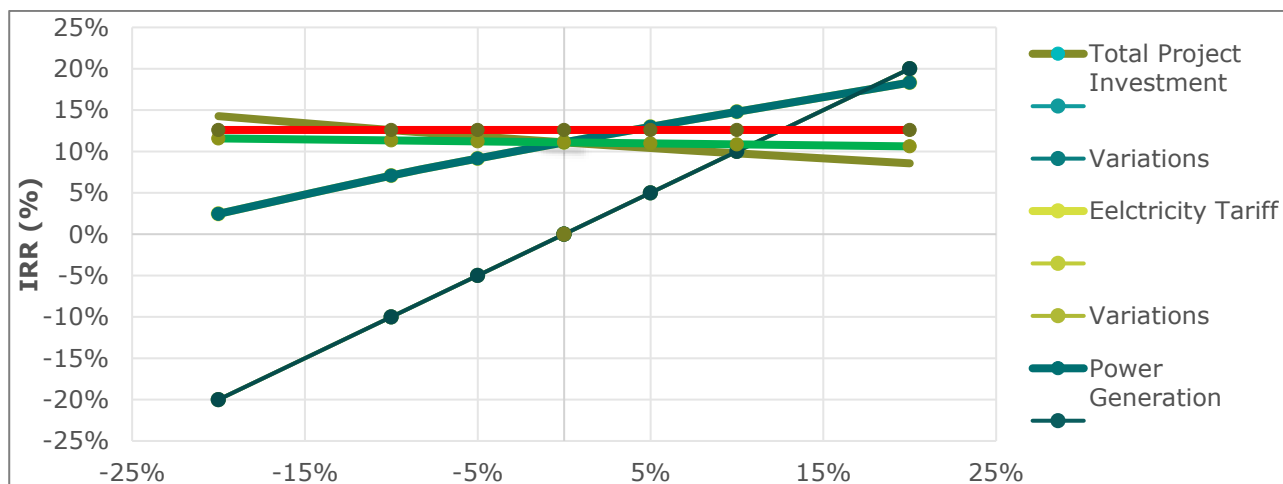
The project investment is already fixed, and the electricity tariff has been already committed with national grid for 20 years. Therefore, both values are not subjected to variation. Hence, only power generation and O&M expenses are subjected to reasonable variation. However, following factors have been considered into sensitivity analysis:

- Total project investment
- Electricity tariff
- Power generation
- Operation & Maintenance (O&M) costs

The results of sensitivity analysis are as follows:

Variations	-20%	-10%	-5%	0%	5%	10%	20%
Total Investment	14.27%	12.58%	11.82%	11.10%	10.42%	9.77%	8.57%

Electricity Tariff	2.46%	7.09%	9.15%	11.10%	12.98%	14.80%	18.33%
Power Generation	2.46%	7.09%	9.15%	11.10%	12.98%	14.80%	18.33%
O&M Costs	11.58%	11.34%	11.22%	11.10%	10.98%	10.86%	10.62%



Variations	Calculated Equity IRR	Benchmark IRR	Required variation to exceed benchmark
Total Investment	11.10%	12.60%	11% decrease
Electricity Tariff			4% increase
Power Generation			4% increase
O&M Costs			64% decrease

Total Project Investment

The equity investment by project owner is 18.99 million USD which is around 37% of the total project investment cost. To assess the additionality of this project, if equity decreased by 11% (2.09 USD), then equity IRR would exceed the benchmark. As per the contracts/agreements in place, relevant purchase orders are already made, thus the equity investment is impossible to vary in the future.

Electricity Tariff

The electricity tariff is extracted from the Power Purchase Agreement (PPA) signed on 10 Dec 2020 with Vietnam Electricity Corporation. The equity IRR is exceeding the benchmark when the electricity tariff increased up to 4% which is not possible for next 20 years as per Section 1 under Article 6 in the PPA.

Power Generation

Increase in power generation would lead to increase in electricity supply to grid and thereby revenue. The equity IRR is exceeding the benchmark, if the power generation increase 4% which is 4.62 MWh of total estimated power generation 115,386 MWh per annum.

The Energy Yield Assessment (EYA) report states that the possible highest net power generation is 95,661 MWh (P50 scenario of wind power generation) which is used to compute the emission reduction in this PDD. The net power generation is excluded all the wake losses (referred in the EYA report) from the gross power generation.

However, we used the gross power generation of 115,386 MWh for the IRR estimation. The gross power generation is already 20% higher than the net power generation and therefore no possibility of power generation increase by 5% in the future. The EYA report was conducted based on the on-site wind measurement (covered minimum 12 calendar months period), equipment specification, potential losses and whether condition as per geographical location. Hence, the possibility of increase in power generation is unlikely in the future.

Operation & Maintenance Costs

The analysis shows that the equity IRR would exceed the benchmark, if the O&M costs decrease more than 60% which is not reasonable. Hence, the O&M costs associated with the project has no impact on the financial returns of the equity.

Step 3: Barrier Analysis

Barrier analysis is not considered.

Step 4: Common practice analysis

Common practice analysis has been carried out as per TOOL24: Common Practice, (Version 03.1). As per the methodological tool, the following definitions are explained in order to evaluate the project as per the stepwise approach for common practice:

Definitions as per methodological tool	Justification
Step 1: calculate applicable capacity or output range as +/-50% of the total design capacity or output of the proposed project activity.	+50% installed capacity : 45 MW Installed capacity : 30.4 MW -50% installed capacity : 15 MW
Step 2: identify similar projects (both CDM and non-CDM) which fulfil all of the following conditions:	Identification of similar projects (CDM & Non-CDM) is carried out as follows: (a) The applicable geographical area is Vietnam (host country) and thereby

(a) The projects are located in the applicable geographical area; (b) The projects apply the same measure as the proposed project activity; (c) The projects use the same energy source/fuel and feedstock as the proposed project activity, if a technology switch measure is implemented by the proposed project activity; (d) The plants in which the projects are implemented produce goods or services with comparable quality, properties and applications areas (e.g. clinker) as the proposed project plant; (e) The capacity or output of the projects is within the applicable capacity or output range calculated in Step 1; (f) The projects started commercial operation before the project design document (CDM-PDD) is published for global stakeholder consultation or before the start date of proposed project activity, whichever is earlier for the proposed project activity.	all projects located in Vietnam are considered (b) The proposed project activity is a green-field renewable energy project and uses measure (b) "Switch of technology with or without change of energy source including energy efficiency improvement as well as use of renewable energies". Therefore, all power generation plants based on renewable energy are considered. (c) The proposed project activity utilizes wind energy source. Hence, only wind energy-based power plants have been considered. (d) The project activity produces renewable electricity by using wind energy source; therefore, all wind power plants are considered. (e) The capacity range of the projects is within the applicable capacity range from 15 to 45 MW (f) The project start date was 30th October 2020. Numbers of Similar projects (CDM and non-CDM) identified, which fulfil above-mentioned conditioned are: In accordance with table shown below, there were 6⁹ wind power plants identified before the investment decision made on 30th October 2020 for the proposed project activity as follows: 1. Bình Thuận 1 (Tuy Phong) – 30 MW 2. Mũi Dinh – 37.6 MW
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⁹ <https://devi-renewable.com/>

	3. Đầm Nại (phase II) – 30 MW 4. Điện gió Hướng Linh 2 – 30 MW 5. Phương Mai 3 – 21 MW 6. Thuận Nhiên Phong – 32 MW
Step 3: within the projects identified in Step 2, identify those that are neither registered CDM project activities, project activities submitted for registration, nor project activities undergoing validation. Note their number N_{all}.	Within the projects identified in Step 2, a total of 3 projects are registered as CDM projects. The remaining projects are as follows: 1. Mũi Dinh – 37.6 MW 2. Đầm Nại (phase II) – 30 MW 3. Điện gió Hướng Linh 2 – 30 MW and therefore, $N_{all} = 3$
Step 4: within similar projects identified in Step 3, identify those that apply technologies that are different to the technology applied in the proposed project activity. Note their number N_{diff}.	As per the definitions provided in Common Practice Analysis tool for the “different technologies”, the project activities have been categorized from the different technologies on the following basis: 12. c) Size of Installation (power capacity)/energy savings Since project activity is large scale project, small and micro scale projects are considered as different technology project. Based on this criterion, there are no different technology project out of the similar identified projects in Step 3. Hence, $N_{diff} = 0$
Step 5: calculate factor $F=1-N_{diff}/N_{all}$ representing the share of similar projects (penetration rate of the measure/ technology) using a measure/ technology similar to	$F=1-N_{diff}/N_{all}$ $F = 1- 0/3 = 1$

the measure/ technology used in the proposed project activity that deliver the same output or capacity as the proposed project activity.	
As per paragraph 18, The proposed project activity is a “common practice” within a sector in the applicable geographical area if the factor F is greater than 0.2 and $N_{all}-N_{diff}$ is greater than 3.	Since F is 1 which is >0.2 however $N_{all}-N_{diff}$ is 3 which is not greater than 3. Hence, the project activity is not a “common practice” in the applicable geographical area.

The above explanation/justification clearly shows that the proposed project activity is unlikely to be financially attractive as well as not a common practice in host country; hence the proposed project activity is additional.

B.5.1 Prior Consideration

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Prior to project start date 30th Oct 2020, an official letter dated on 01st Sep 2020 was circulated to all the board members by the chairperson. The letter describes the potential revenue generation through carbon credits and the process to obtain them. The board was planning to cover the O&M costs through the carbon revenue. Its clear that the project owners considered the carbon revenue prior project implementation.

The chronology of project implementation and GS certification as follows:

Date	Significant Actions
30/10/2020	Date of EPC (Engineering, Procurement & Construction) contract signed – start date of the project
23/11/2020	Authority’s approval on the basic design
24/11/2020	SCADA Agreement
10/12/2020	Power Purchase Agreement (PPA) name transfer
28/01/2021	Measurement & Verification Agreement with EVN
10/09/2021	Submission to GS for project listing
29/10/2021	Expected commercial operation
28/10/2021	First submission for the GS preliminary review
29/03/2022	Completion of GS preliminary review

B.5.2 Ongoing Financial Need

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Not applicable

B.6. Sustainable Development Goals (SDG) outcomes

Relevant Target/Indicator for each of the three SDGs

Sustainable Development Goals Targeted	Most relevant SDG Target	SDG Impact Indicator (Proposed or SDG Indicator)
13 Climate Action (mandatory)	Target 13.3: Integrate climate change measures into national policies, strategies grid planning	Indicator 13.2.2. GHG emission reductions per year
8 Decent Work and Economic Growth	Target 8.5: By 2030, achieve full and productive employment and decent work for all women and men, including for young people and persons with disabilities, and equal pay for work of equal value	Indicator 8.5.1 – Average hourly earnings of female and male employees, by occupation, age and persons with disabilities
7 Affordable and Clean Energy	Target 7.2: By 2030, increase substantially the share of renewable energy in the global energy mix	Indicator 7.2.1 – Renewable energy share in the total final energy consumption (MWh of electricity generation)

B.6.1 Explanation of methodological choices/approaches for estimating the SDG Impact

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SDG	Methodological Approaches	QA/QC
SDG 13: Climate Action	The emission reduction parameter is calculated through net electricity supplied to grid and grid emission factor. The grid emission factor is published (CV 263/BDKH ¹⁰) by Department of Climate Change - Ministry of Natural Resources	As per the chosen methodology, equations: (11): $BE_y = EG_{pj,y} \times EF_{grid,CM,y}$ (17): $ER_y = BE_y \times PE_y$

¹⁰ [http://dcc.gov.vn/van-ban-phap-luat/1059/Nghien-cuu-xay-dung-heso-phat-thai-\(EF\)-cua-luoi-dien-Viet-Nam-\(K%C3%A8m-CV-263/BDKH\).html](http://dcc.gov.vn/van-ban-phap-luat/1059/Nghien-cuu-xay-dung-heso-phat-thai-(EF)-cua-luoi-dien-Viet-Nam-(K%C3%A8m-CV-263/BDKH).html)

	and Environment and explained in section B.6.2. This is in line with "Tool to calculate the emission factor for an electricity system, version 7". The emission reductions are calculated as per the chosen methodology in section B.6.3.	Where: BE = Baseline emission EG = Power generation EF = Grid emission factor ER = Emission reduction PE = Project emission were used to calculate the baseline emission & emission reduction and it is further explained in section B.6.4 as well as used in emission reduction spreadsheet. This parameter is calculated and thereby no QA/QC procedure is required.
SDG 8: Decent Work and Economic Growth	The average hourly earnings of female/male employees, by occupation, age and persons with disabilities has been checked through employment contract, payslip and payroll records. Randomly choose employees with designation from payroll records and compare their wages, gender, experience and abilities.	This parameter is not required any QA/QC procedure. The DOE can confirm this parameter with review employee's records and interview with project owner or employees on site
SDG 7: Affordable and Clean Energy	Electricity produced and supplied to the grid is monitored through energy meters. A SCADA (Supervisory control and data acquisition) in place to control and monitor the plant operation. Net electricity will be calculated by project owner on monthly basis and provided in the monthly/quarterly/biyearly/yearly report or equivalent. The other parameters used to reach net electricity supplied to grid are mentioned in monitoring plan.	This parameter is monitored monthly basis and value of parameter will be cross checked with invoices to national grid. The electricity meters will be verified by either EVN or calibrated as per supplier recommendations.

B.6.2 Data and parameters fixed ex ante

SDG13

Data/parameter	EF_{grid,OM,y}
Unit	tCO ₂ /MWh
Description	Operating Margin CO ₂ emission factor for the electricity system in year y
Source of data	http://dcc.gov.vn/van-ban-phap-luat/1059/Nghien-cuu,-xay-dung-heso-phat-thai-(EF)-cua-luoi-dien-Viet-Nam-(K%C3%A8m-CV-263/BDKH).html
Value(s) applied	0.8795
Choice of data or Measurement methods and procedures	The latest EF _{grid,OM,y} for 2018 was calculated and published (CV 263/BDKH) by Department of Climate Change - Ministry of Natural Resources and Environment on 12/03/2020 as per "Tool to calculate the emission factor for an electricity system (Version 07.0)"
Purpose of data	To compute combined margin CO ₂ emission factor for the electricity system
Additional comment	None

SDG13

Data/parameter	EF_{grid,BM,y}
Unit	tCO ₂ /MWh
Description	Build Margin CO ₂ emission factor for the electricity system in year y
Source of data	http://dcc.gov.vn/van-ban-phap-luat/1059/Nghien-cuu,-xay-dung-heso-phat-thai-(EF)-cua-luoi-dien-Viet-Nam-(K%C3%A8m-CV-263/BDKH).html
Value(s) applied	0.9465
Choice of data or Measurement methods and procedures	The latest EF _{grid,BM,y} for 2018 was calculated and published (CV 263/BDKH) by Department of Climate Change - Ministry of Natural Resources and Environment

	on 12/03/2020 as per "Tool to calculate the emission factor for an electricity system (Version 07.0)"
Purpose of data	To compute combined margin CO ₂ emission factor for the electricity system
Additional comment	None

SDG13

Data/parameter	EF_{grid,CM,y}
Unit	tCO ₂ /MWh
Description	Combined Margin CO ₂ emission factor for the electricity system in year y
Source of data	http://dcc.gov.vn/van-ban-phap-luat/1059/Nghien-cuu,-xay-dung-heso-phat-thai-(EF)-cua-luoi-dien-Viet-Nam-(K%C3%A8m-CV-263/BDKH).html
Value(s) applied	0.89625
Choice of data or Measurement methods and procedures	The EF_{grid,CM,y} is calculated by using Operating Margin (EF_{grid,OM,y}) and Build Margin (EF_{grid,BM,y}) CO₂ emission factors published (CV 263/BDKH) by Department of Climate Change - Ministry of Natural Resources and Environment on 12/03/2020 as per "Tool to calculate the emission factor for an electricity system (Version 07.0)". the computation as follows: $EF_{grid,CM,y} = (EF_{grid,OM,y} \times W_{OM}) + (EF_{grid,BM,y} \times W_{BM})$ Where: W_{OM} = Weighting of OM emissions factor (%) = 75% W_{BM} = Weighting of BM emissions factor (%) = 25%
Purpose of data	To compute baseline emission for greenfield grid connected wind power plants
Additional comment	None

B.6.3 Ex ante estimation of SDG Impact

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SDG 7: Affordable and Clean Energy

The baseline outcome benefit is zero.

The project activity generates 95,661¹¹ MWh which will be monitored by electricity meters. No ex-ante estimation required for SDG 7.

SDG 8: Decent Work and Economic Growth

The baseline outcome benefit is zero.

The project provided an employment opportunity for 11 local people (10 men and 1 women) during the construction phase and they are planning to employ more local people permanently during operational phase. It may vary according to the operation needs throughout project lifetime. This will help to achieve SDG 8 with indicators 8.5.2 "Unemployment rate, by sex, age and persons with disabilities" and following target: 8.5 "By 2030, achieve full and productive employment and decent work for all women and men, including for young people and persons with disabilities, and equal pay for work of equal value". Number of jobs created and wages for both men and women to demonstrate that the project provides decent work for all women and men and equal pay for equal work. The pay for men and women will be proportional to nature of work which they engaged for and the working hours is similar to all employees. This will be monitored through payroll records and attendance system which is available at the project site.

SDG 13: Climate Action

Baseline Emission:

As per Equation (11) of ACM0002 (Version 20.0), the baseline emissions are to be calculated as follows:

$$BE_y = EG_{PJ,y} \times EF_{grid,CM,y}$$

Where:

BE_y = Baseline emissions in year y (tCO₂/yr)

$EG_{PJ,y}$ = Quantity of net electricity generation that is produced and fed into the grid as a result of the implementation of the CDM project activity in year y (MWh/yr)

¹¹ The net power generation is estimated through page 74 of feasibility report.

$EF_{grid,CM,y}$ = Combined margin CO₂ emission factor for grid connected power generation in year y calculated using the latest version of "TOOL07: Tool to calculate the emission factor for an electricity system" (tCO₂/MWh)

As per Equation (12) of ACM0002 (Version 20.0), the $EG_{PJ,y}$ for the greenfield power plant is calculated as follows:

$$EG_{PJ,y} = EG_{facility,y}$$

$EG_{facility,y}$ = Quantity of net electricity generation supplied by the project plant/unit to the grid in year y (MWh/yr)

The est. electricity generation¹² by the proposed project ($EG_{PJ,y}$) = 95,661 MWh/yr

The CM emission factor for the Vietnam's electricity system ($EF_{grid,CM,y}$) = 0.89625 tCO₂/MWh.

Hence the baseline is:

$EG_{PJ,y}$	×	$EF_{grid,CM,y}$	=	BE_y
95,661 MWh/yr	×	0.89625 tCO ₂ /MWh	=	85,736 tCO ₂ /yr

Project Emission:

The proposed project activity involves the generation of electricity by development of wind power project. The generation of electricity does not result in greenhouse gas emissions and therefore the project emission (PE_y) is zero.

Emission Reduction:

As per Equation (17) of ACM0002 (Version 20.0), the emission reduction (ER_y) of the proposed project as follows:

BE_y	–	PE_y	=	ER_y
85,736 tCO ₂ /yr	–	0 tCO ₂ /yr	=	85,736 tCO ₂ /yr

B.6.4 Summary of ex ante estimates of each SDG Impact

SDG 7: Affordable and Clean Energy – Ensure access to affordable, reliable, sustainable and modern energy for all

¹² Estimated (for 1st year) by using power simulations variant – Done by ERS Energy Sdn Bhd (Malaysia)

Year	Baseline estimate (MWh)	Project estimate (MWh)	Net benefit (MWh)
2021 (2 months only)	0	15,944	15,944
2022	0	95,661	95,661
2023	0	95,661	95,661
2024	0	95,661	95,661
2026	0	95,661	95,661
2027 (10 months only)	0	79,717	79,717
Total	0	478,305	478,305
Total number of crediting years	5		
Annual average over the crediting period	0	95,661	95,661

SDG 8: Decent Work and Economic Growth – Promote inclusive and sustainable economic growth, employment and decent work for all

The project leads to employment opportunities which would not have been possible in the baseline scenario. The project provided an employment opportunity of 11 local people (10 men and 1 women) during the construction phase and they are planning to employ more local people permanently during operational phase.

SDG 13: Climate Action – Take urgent action to combat climate change and its impacts

Year	Baseline estimate (tCO ₂ e)	Project estimate (tCO ₂ e)	Net benefit (tCO ₂ e)
2021 (2 months only)	14,289	0	14,289
2022	85,736	0	85,736
2023	85,736	0	85,736
2024	85,736	0	85,736
2026	85,736	0	85,736
2027 (10 months only)	71,447	0	71,447
Total	428,680	0	428,680
Total number of crediting years	5		

Annual average over the crediting period	85,736	0	85,736
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B.7. Monitoring plan

B.7.1 Data and parameters to be monitored

SDG 13: Climate Action

Data / Parameter	ER _y - Annual emission reduction
Unit	tCO ₂ e/year
Description	The project will generate renewable energy, which will displace part of the electricity otherwise supplied by fossil fuel fired power plants, thereby the emission reductions are achieved by this project
Source of data	Calculated by net electricity generated by the project activity and combined margin emission factor of electricity system.
Value(s) applied	85,736 tCO ₂ e/year
Measurement methods and procedures	Computed as recommended by the ACM000 (Version 20.0)
Monitoring frequency	Once for each monitoring period
QA/QC procedures	This parameter is cross checked with the electricity supplied to grid
Purpose of data	Direct emission reductions of the Project over the Baseline
Additional comment	None

SDG 8: Decent Work and Economic Growth (Indicator 8.5.1 – Average hourly earnings of female and male employees, by occupation, age and persons with disabilities)

Data / Parameter	Employees' monthly wages
Unit	Vietnam Dong per employee
Description	Monthly wages to all employees by the project activity
Source of data	Job contracts (or) Pay slip (or) Payroll records

Value(s) applied	Employees' wages record available during validation.
Measurement methods and procedures	Randomly choose employees with similar designation from payroll records and compare their wages and experience
Monitoring frequency	Once for each monitoring period
QA/QC procedures	Payroll records, job contracts and staff register will be provided annually. After first verification, only changes in employees will be reported.
Purpose of data	To demonstrate contribution to SDG8- 8.5 "By 2030, achieve full and productive employment and decent work for all women and men, including for young people and persons with disabilities, and equal pay for work of equal value'
Additional comment	None

SDG 7: Affordable and Clean Energy (Indicator 7.2.1: Renewable energy share in the total final energy consumption)

Data / Parameter	$EGP_{J,grid,y}$
Unit	MWh/yr
Description	Net electricity supplied to national grid by the project activity
Source of data	Direct measurement by energy meters at point of electricity supplied to the national grid
Value(s) applied	95,661 MWh/year (this value may change in future due to fluctuation in power generation)
Measurement methods and procedures	Electricity generated would be monitored by the electricity meters on a continuous basis and readings will be taken every month for billing purpose. The meters have the capability to store data as well as transfer them to server which is located at the control room in switch station. It can be downloaded on a monthly basis for billing purpose. Archive of the data would be available for entire lifetime of the project.
Monitoring frequency	Continuous measurement with daily recording

	(The meters can capture half hourly generation data and the data would be logged through the SCADA system in a daily basis. However, readings will be downloaded on a monthly basis for record keeping and billing to the EVN.)
QA/QC procedures	Electricity meters will be calibrated once in three years as per EVN requirement. The electricity generation can also be cross-checked using the invoices billed to EVN EPTC for payment
Purpose of data	Calculation of baseline emissions to demonstrate contribution to SDG7'-7.2 By 2030, increase substantially the share of renewable energy in the global energy mix"
Additional comment	None

Safeguarding Principle Assessment

Data / Parameter	3. Community Health, Safety and Working Conditions
Unit	Not applicable
Description	The Project shall avoid community exposure to increased health risks and shall not adversely affect the health of the workers and the community.
Source of data	Training records (either attendance register or event photos of the training)
Value(s) applied	Not applicable
Measurement methods and procedures	The training records include attendance registers and photos will be available to measure the level of safety trainings conducted for workers and awareness training provided to community regarding climate change and mitigation measures.
Monitoring frequency	Once for each monitoring period
QA/QC procedures	Cross checking by interviews
Purpose of data	Improve employees' health and safety
Additional comment	None

B.7.2 Sampling plan

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Not applicable.

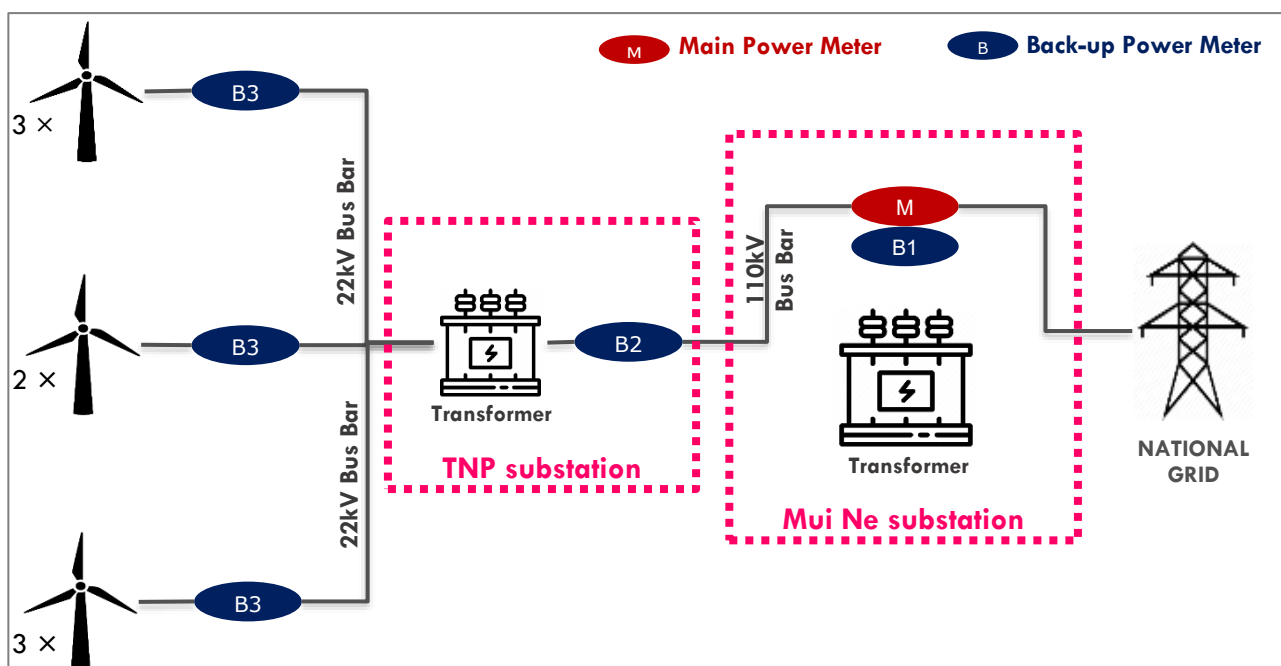
B.7.3 Other elements of monitoring plan

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Monitoring data is collected in accordance with the agreement done between the project owner and Vietnam Electricity Corporation (EVN) which provides the infrastructure for the connection to the national grid. Data will be stored electronically, during the crediting period and at least two years after the last issuance of credits for the solar power project activity in the concerning crediting period. The Project Participant will be responsible for storage of data received from the measuring devices.

Monitoring procedures

The meters (highlighted in red) are installed as follows:



All the installed electricity meters have an internal memory that can be stored monitored data. The electricity meters will obtain half hourly data and send to the server located at control room in switch station on a daily basis. The data can be downloaded from the computer at control room.

Period of archiving

The data will be archived electronically every half hour in meters and the meters will send the data to server on a daily basis. Monthly data can be downloaded for billing purposes.

Calibration of equipment

A total of 6 electricity meters (1 main & 5 backup) installed various location of the plant. The main meter is installed at Mui Ne station where 3 other power plants connected with. The calibration of electricity meters will be done as per supplier recommendation by an independent accredited third party. However, the main meter is validated by EVN as per their requirement. The relevant calibration certificates will be made available at the time of verification. The technical specification of the meter as follows:

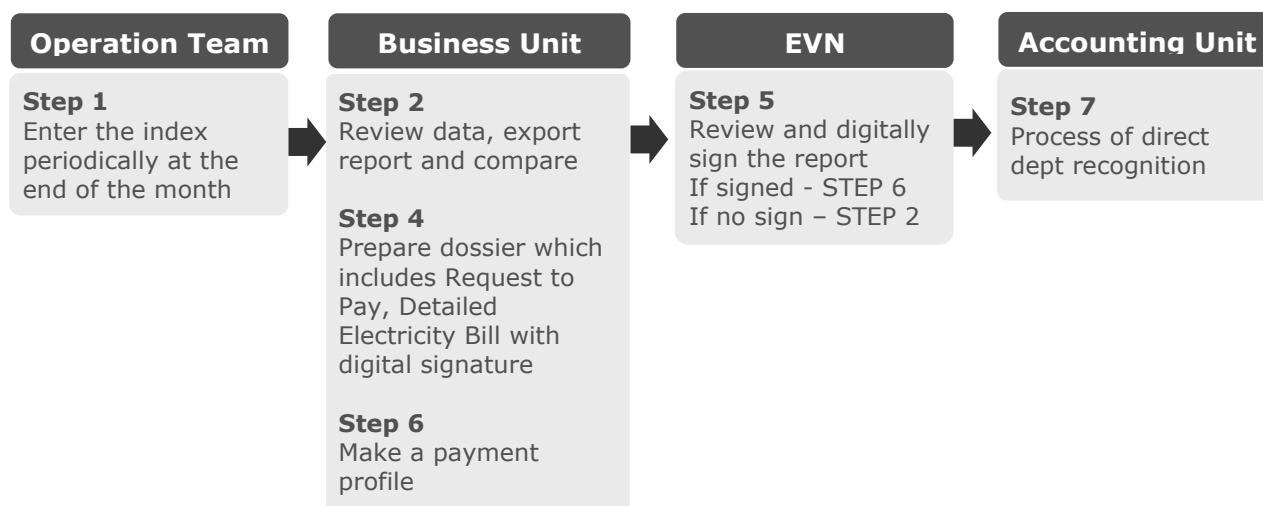
Parameters	Description (or) Value	
	Main Meter	Back-up Meter
Model	Landis+Gyr E650 (ZMD402C)	Landis+Gyr E650 (ZMD405C)
Accuracy level	Class 0.2	Class 0.5
Application	High Voltage	
Connection Type	Transformer connection	
Energy Type	Active/Reactive Energy	

Training requirements

The suppliers of the equipment will train the staff in-charge during erection, to operate and maintain the equipment efficiently. Apart from this, the equipment supplier will provide complete manuals and documentation providing details for the maintenance schedule and the required activities associated with it.

Roles & Responsibilities

The following roles and responsibilities will be used to monitor the project activity and invoicing to grid as per procedures:



The monitored data will be reported on a yearly basis for the calculation and estimation of emission reductions. This data will be checked against the billing to EVN. If the project is not performing as expected or if there are any negative impacts on the volume of emission reductions obtained, identify the where the project is deviating in its generation of emission reductions and the immediate measures which need to be undertaken to maintain the expected generation of emission reductions from the operation of this project. Should there be any significant changes in plant operation, these will be notified and amendments to the PDD will be requested during the following verification by GS-VVB.

For each verification period, project owner will prepare a monitoring report that will be submitted to a GS-VVB for verification. Project owner ensures that the procedures and monitoring plan are being followed. All data will be kept for a minimum of 2 years following issuance of certified emission reductions or the end of the crediting period, whichever is later, and the storage of this data will be the responsibility of the project owners.

SECTION C. DURATION AND CREDITING PERIOD

C.1. Duration of project

C.1.1 Start date of project

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30 October 2020¹³

C.1.2 Expected operational lifetime of project

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25 years¹⁴

C.2. Crediting period of project

C.2.1 Start date of crediting period

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29 October 2021 (first crediting period – CP1)

C.2.2 Total length of crediting period

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CP1 – 5 years 0 months

CP2 – 5 years 0 months

CP3 – 5 years 0 months

SECTION D. SUMMARY OF SAFEGUARDING PRINCIPLES AND GENDER SENSITIVE ASSESSMENT

D.1 Safeguarding Principles that will be monitored

A completed Safeguarding Principles Assessment is in [Appendix 1](#), ongoing monitoring is summarised below.

Principles	Mitigation Measures added to the Monitoring Plan
Principle 3 – Community Health, Safety and Working Conditions	The training records include attendance registers and photos will be available to measure the level of safety trainings conducted for workers and awareness training provided to community regarding climate change and mitigation measures. For more details, please refer section B.7.1

¹³ Date of Engineering, Procurement & Construction (EPC) contract signed

¹⁴ As per "Option (c)" of "TOOL10: Tool to determine the remaining lifetime of the equipment (Version 01)"

Principle 6.1 – Human Rights	Appendix 3 – As per the labour code and regulations, project owner has the necessary documents to demonstrate the compliance towards: 1. Working hours (must not exceed 48 hours per week on a regular basis), AND 2. Duties and tasks, AND 3. Remuneration (must include provision for payment of overtime), AND 4. Modalities on health insurance, AND 5. Modalities on termination of the contract with provision for voluntary resignation by employee, AND 6. Provision for annual leave of not less than 10 days per year, not including sick and casual leave. through employment contracts and employee insurance policies. Hence, there is no need to include separate table under monitoring plan.
Principle 9.4 – Release of Pollutants	For noise: Inspection of the noise level of works should be carried out at all times; and Reporting of any excessive levels of noise and the measures taken to minimize the impact and prevent from occurring again. For flickering: Monitoring and reporting of any complains/grievances on shadow flicker effect during operational phase so as appropriate mitigation measures can be assessed and provided to the sensitive receptors.
Principle 9.5 – Hazardous and Non-Hazardous Waste	▪ Inspection of hazardous waste management practices onsite. ▪ Review of records and manifests for volume of hazardous waste generated to ensure consistency; and ▪ Regular environmental reporting on implementation of the hazardous waste management practices onsite, including reporting of the hazardous waste management to local DoNRE during the construction (within one month from completion of construction). ▪ During operational phase, the domestic waste generation is very minimum and the waste is disposed by the third-party waste collector.

Principle 9.10 – High Conservation Value Areas and Critical Habitats

A carcass monitoring program is to be conducted on a weekly basis at the base of all turbines.
A review of the data collected from carcasses monitoring is to be undertaken every 6 months for the first 2 years. Wind farm operations may be altered based on the lifecycle characteristics of any species identified that are susceptible to strike. Assessment of data is to occur yearly from the 3rd year of operation.

D.2. Assessment that project complies with GS4GG Gender Sensitive requirements

Question 1 - Explain how the project reflects the key issues and requirements of Gender Sensitive design and implementation as outlined in the Gender Policy?	The project owner has prepared the project activity to ensure the contribution to generate livelihood benefits and enable women and men to equitably contribute to the local development through the project implementation.
Question 2 - Explain how the project aligns with existing country policies, strategies and best practices	The project activity totally complies with the Gender Equality Law approved by the National Assembly in 2006 and the Government's Decree No. 48/2009/ND-CP providing measures to assure gender equality.
Question 3 - Is an Expert required for the Gender Safeguarding Principles & Requirements?	No. The Project Owner has its own internal regulations including their commitments to implement the gender equality. All the GS mandatory requirements of the gender equality and women's right are met properly.
Question 4 - Is an Expert required to assist with Gender issues at the Stakeholder Consultation?	To be confirmed upon completion of stakeholder consultation. Anyhow, project owner is planning 3 different consultation meetings and "women only" meeting one of them.

SECTION E. SUMMARY OF LOCAL STAKEHOLDER CONSULTATION

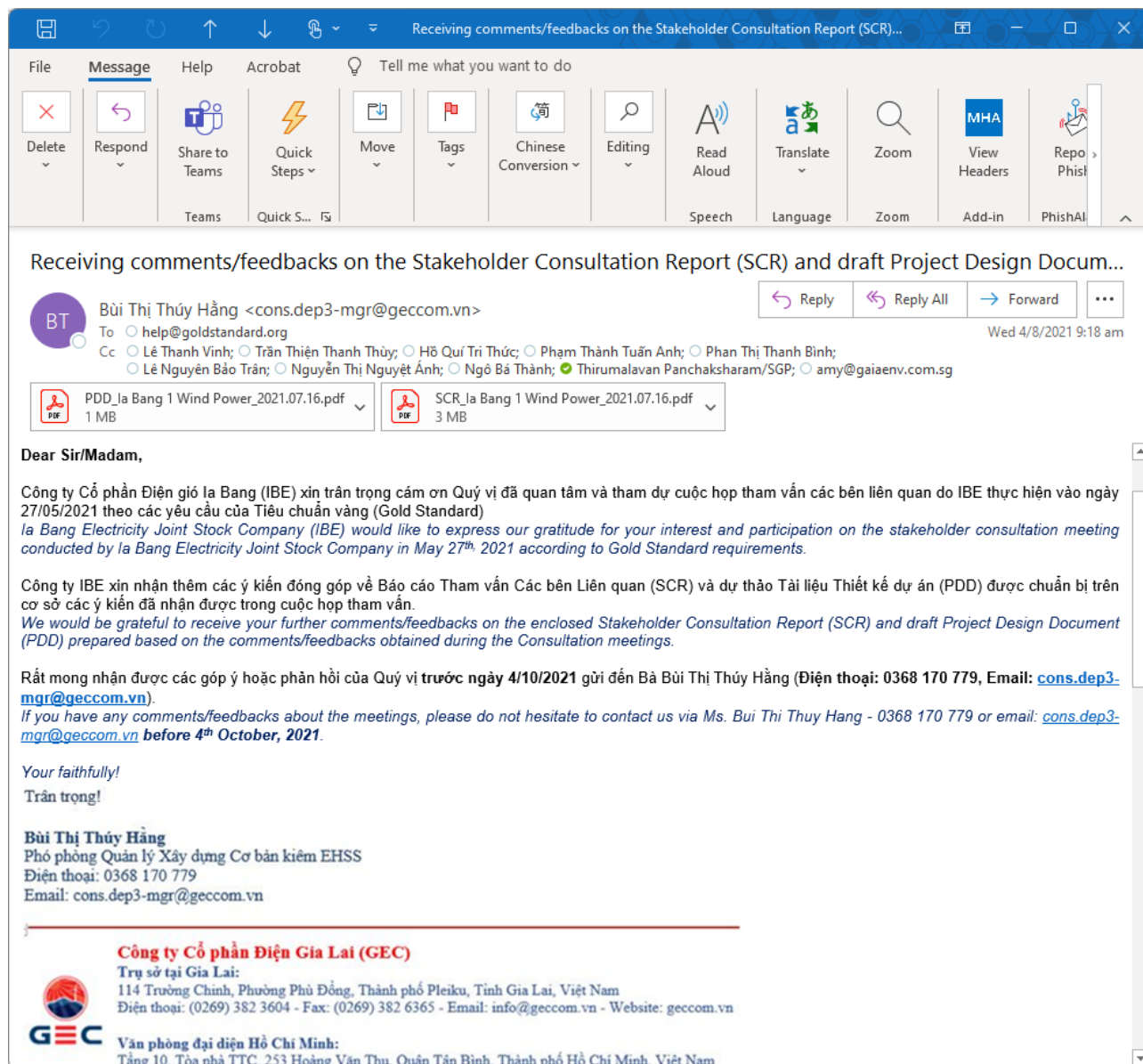
The below is a summary of the 2 step GS4GG Consultation for monitoring purposes. Please refer to the separate Stakeholder Consultation Report for a complete report on the initial consultation and stakeholder feedback round.

E.1 Summary of stakeholder mitigation measures

>>

The stakeholder meeting was organized on 28th June 2021 (from 9:30 onwards). The gathering at Hoa Thang commune People's committee

meeting hall is prohibited in accordance with social distance rule due to pandemic spread in the area. Hence, multiple small group (not more than 5 pax) meetings were conducted at the meeting hall. A total of 43 stakeholders attended the meetings. The feedback round was carried out for 60 days (from 04/08/2021 to 04/10/2021).



Gender of Stakeholder	Stakeholder comment	Was comment taken into account (Yes/ No)?	Explanation (Why? How?)
Male	What is the initial capacity of the TNP	Yes	The capacity of the project according to the installation plan

	project and how is it expected to generate electricity to serve the lives of people in the area?		will reach 30 MW with an estimated annual electricity output of 95,661,000 kWh per year. The project contributes to the electricity demand of Hoa Thang commune, Bac Binh district.
Female	The weather in Binh Thuan is usually very dry, with a lot of sunshine and little wind, is it effective to invest in wind power projects	Yes	Based on calculations, the annual wind volume at TNP's location has a very good wind index, ensuring the exploitation of wind resources at an average wind level. In addition, the location right in the strait also helps the wind turbines to catch sea breezes very well. The dry time of the project is only in the days of April-June, most of it is windy, but the wind volume is not as high as the rest of the months. In general, in Binh Thuan, the investment in wind power may not be feasible but contributes a lot to the economy and living environment in the future.
Male	Does the investment in wind power with such a large cost affect the environment, the lives of the surrounding people and have any benefits to society?	Yes	Wind power projects in general and TNP wind power project in particular bring great economic value and contribute to the protection of the living environment. With advantages such as: ▪ Use less land resources ▪ The energy produced is evenly distributed over the years, avoiding overloading of transmission lines ▪ Use natural energy is wind

- Does not cause adverse effects on geology, does not affect soil erosion and desertification.

That's why we decided to invest in wind power projects.

Male	Are households near the project area affected by noise or living environment?	Yes	The level of impact by noise will not be significant, because the project is planned to separate it from the residential area, in order to limit the above problem.
Female	What is Golden Standard?	No	GS is the most widely respected certification standard in global level for carbon offset projects. It ensures those carbon offset projects actually provide benefits to the local population. Such benefits include positive impacts on the well-being of the local community.

E.2 Final continuous input / grievance mechanism

Method	Include all details of Chosen Method (s) so that they may be understood and, where relevant, used by readers.
Continuous Input / Grievance Expression Process Book (mandatory)	Grievance expression process book The grievance expression process books have been located at the site office (Address: Hoa Thang commune, Bac Binh district, Binh Thuan province, Vietnam) and corporate office (344 Phan Chau Trinh, Binh Thuan, Hai Chau, Da Nang) which can be access by public. The location and purpose of process book was clearly explained during the meeting. Project Participant will be checking the comments in the book on a regular basis, and record responses. They will be respectful to the views of stakeholders and suggest alternative solutions

or compromises whenever possible. It was chosen as a main mechanism by stakeholders.

GS Contact (mandatory)	help@goldstandard.org
Telephone access (optional)	Mr Triết Phạm Đỗ Minh (Chief Accountant) Mobile: +84 236 3562 361 This information has been explained at the consultation meeting and provided in the Continuous Input/Grievance Expression Process Book and invitation.
Internet/email access (optional)	Mr Triết Phạm Đỗ Minh (Chief Accountant) E-mail: info@vneco.com.vn This information has been explained at the consultation meeting and provided in the Continuous Input/Grievance Expression Process Book and invitation. It is chosen as main mechanism.

APPENDIX 1 - SAFEGUARDING PRINCIPLES ASSESSMENT

Complete the Assessment below and copy all Mitigation Measures for each Principle into [SECTION D](#) above. Please refer to the instructions in the [Guide to Completing](#) this Form.

Assessment Questions/ Requirements	Justification of Relevance (Yes/ potentially/ no)	How Project will achieve Requirements through design, management or risk mitigation.	Mitigation Measures added to the Monitoring Plan (if required)
Principle 1. Human Rights			
1. The Project Developer and the Project shall respect internationally proclaimed human rights and shall not be complicit in violence or human rights abuses of any kind as defined in the Universal Declaration of Human Rights 2. The Project shall not discriminate with regards to participation and inclusion	No	1. The Project Proponent and the Project does not conflict with the economic livelihood or other factors of the local communities. Thus, the Project does not cause any human rights abuse and respects internationally proclaimed human rights issue. 2. As a member of United Nations and part of UN Agreement on Human Rights¹⁵, it is ensured by	Not required

¹⁵ <https://thuvienphapluat.vn/van-ban/Quyen-dan-su/Tuyen-ngon-quoc-te-nhan-quyen-1948/65774/noi-dung.aspx>

		law in Vietnam that no action can be taken against human rights.	
Principle 2. Gender Equality			
1. The Project shall not directly or indirectly lead to/contribute to adverse impacts on gender equality and/or the situation of women 2. Projects shall apply the principles of nondiscrimination, equal treatment, and equal pay for equal work 3. The Project shall refer to the country's national gender strategy or equivalent national commitment to aid in assessing gender risks 4. (where required) Summary of opinions and recommendations of an Expert Stakeholder(s)	No	1. The project will benefit to local residents regardless of gender and it does not create any adverse effect on the local community. Also, it does not discriminate the local community on basis of gender or caste or religion and therefore equally serve to all. Hence, it will not expose women and girls to further risks or hazards. 2. The Project has equal opportunity for women and men to contribute both in volunteer and working positions. Project owner has a specified HR policy that considers participation by both men and women. 3. The project does not involve in any form of discrimination in any	Not required

		kind. Vietnam also ratified 16 relevant ILO core conventions on equality, namely Equal Remuneration Convention (Convention No 100) and Discrimination (Employment and Occupation) Convention (Convention No 111) in 1997. 4. Not required as government bodies who are expert in their field were attended the meeting and there were no comments from them.	
Principle 3. Community Health, Safety and Working Conditions			
1. The Project shall avoid community exposure to increased health risks and shall not adversely affect the health of the workers and the community	Yes	1. The project proponent is committed to the employee's workplace health & safety during all phases of the project. All employees will attend health & safety trainings. This issue is	Workplace Health & Safety trainings will be conducted regularly during the project operation. This is a monitoring

¹⁶ https://www.ilo.org/dyn/normlex/en/f?p=1000:11200:0::NO:11200:P11200_COUNTRY_ID:103004

		protected by Labor code ¹⁷ and UN Agreement on Human Rights ¹⁸ .	parameter under section B.7.1
Principle 4.1 Sites of Cultural and Historical Heritage			
Does the Project Area include sites, structures, or objects with historical, cultural, artistic, traditional or religious values or intangible forms of culture?	No	The Project Site does not have any physical cultural heritage. Consultation with the local communities also indicated that the Site is unlikely to have any significant underground archaeological or cultural remains. Law on Cultural heritage is protected against alteration, damage or removal by the “law on cultural heritage” ¹⁹ .	Not required
>>			
Principle 4.2 Forced Eviction and Displacement			
Does the Project require or cause the physical or economic relocation of peoples (temporary or permanent, full or partial)?	No	The project does not involve any settlement areas. Thus, this project does not cause the physical or economic relocation of peoples. The project activity does not involve any alteration of existing lifestyle of residents. Since	Not required
>>			

¹⁷ <https://thuvienphapluat.vn/van-ban/Lao-dong-Tien-luong/Bo-Luat-lao-dong-2012-142187.aspx>

¹⁸ <https://www.ohchr.org/EN/Countries/AsiaRegion/Pages/VNIndex.aspx>

¹⁹ <https://thuvienphapluat.vn/van-ban/Van-hoa-Xa-hoi/Van-ban-hop-nhat-10-VBHN-VPQH-2013-hop-nhat-Luat-di-san-van-hoa-204826.aspx>

		the site located isolated area, no additional burden to the residents.	
Principle 4.3 Land Tenure and Other Rights			
a. Does the Project require any change, or have any uncertainties related to land tenure arrangements and/or access rights, usage rights or land ownership? b. For Projects involving land use tenure, are there any uncertainties with regards to land tenure, access rights, usage rights or land ownership?	Yes	a. TNP is leased the land from authority and the tenure is 50 years b. No uncertainties. The authority issued "Land Use Rights Certificates" for the land leased by the project owner. The tenure is fixed until July 2064.	Not required
Principle 4.4 - Indigenous people			
Are indigenous peoples present in or within the area of influence of the Project and/or is the Project located on land/territory claimed by indigenous peoples?	No	The project's impact will not affect the distinct habitats or ancestral territories and natural resources. There will be no land acquisition from the indigenous people, and the land that the project occupies are either owned by authority.	Not required
Principle 5. Corruption			
1. The Project shall not involve, be complicit in or inadvertently contribute to or reinforce corruption or corrupt Projects	No	1. The project does not involve and is not complicit in any kind of	Not required

			corruption. Vietnam has ratified ²⁰ UN convention against Corruption in 2007.	
Principle 6.1 Labour Rights				
Appendix 1.	The Project Developer shall ensure that all employment is in compliance with national labour occupational health and safety laws and with the principles and standards embodied in the ILO fundamental conventions	Yes	1. TNP will not involve in any form of forced or compulsory labour. Vietnam has ratified ILO "C029 Forced Labour Convention" ²¹	Appendix 3 – As pre the labour code and regulations, project owner has the necessary documents to demonstrate the compliance towards: 7. Working hours (must not exceed 48 hours per week on a regular basis), AND 8. Duties and tasks, AND 9. Remuneration (must include
Appendix 2.	Workers shall be able to establish and join labour organisations		2. Workers should have the right to establish and join the organization that they consider necessary in a climate of complete security based on Decree No. 8/2014/ND-CP.	
Appendix 3.	Working agreements with all individual		3. The project owner follows regulations of Labour Code of Viet Nam. Thus, they always ensure the participation of women and men in project activities and benefits. The employees of the project are trained on technical aspects relating to the operation of the wind power plant and provided with labour contracts,	

²⁰ https://treaties.un.org/Pages/ViewDetails.aspx?src=IND&mtdsg_no=XVIII-14&chapter=18&clang=en#EndDec

²¹ https://www.ilo.org/dyn/normlex/en/f?p=1000:11200:0::NO:11200:P11200_COUNTRY_ID:103004

workers shall be documented and implemented and include: 1. Working hours (must not exceed 48 hours per week on a regular basis), AND 2. Duties and tasks, AND 3. Remuneration (must include provision for payment of overtime), AND 4. Modalities on health insurance, AND 5. Modalities on termination of the contract with provision for voluntary resignation by employee, AND 6. Provision for annual leave of not less than 10 days per year, not including sick and casual leave. Appendix 4. No child labour is allowed (Exceptions		medical insurance and regular health-check as well as social insurance and unemployment insurance. (Ref. Labour Code). 4. TNP is complying with all relevant national laws regarding child labor. TNP will not employ children in any shape or form for their works. Vietnam has ratified ILO "C138 – Minimum Age Conventions" and "C182 – Worst Forms of Child Labour Convention"²² 5. The project does not require labour force for implementation of the project. Trained technicians are involved in construction, operation and maintenance of plant.	provision for payment of overtime), AND 10. Modalities on health insurance, AND 11. Modalities on termination of the contract with provision for voluntary resignation by employee, AND 12. Provision for annual leave of not less than 10 days per year, not including sick and casual leave. through employment contracts and employee insurance policies.
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²² https://www.ilo.org/dyn/normlex/en/f?p=1000:11200:0::NO:11200:P11200_COUNTRY_ID:103004

Appendix 5.	for children working on their families' property requires an Expert Stakeholder opinion) The Project Developer shall ensure the use of appropriate equipment, training of workers, documentation and reporting of accidents and incidents, and emergency preparedness and response measures			Hence, there is no need to include separate table under monitoring plan.
Principle 6.2 Negative Economic Consequences				
1. Does the project cause negative economic consequences during and after project implementation?	No	The project doesn't cause any negative economic consequences during and after project implementation. Instead, the project gives a boost to the wind energy sector and push R&D in this field thus improving national economy through innovations. Also, contributes to poverty alleviation through employment opportunities during project construction & operation as well as towards nation's tax revenues.	Not required	
>>				

Principle 7.1 Emissions			
Will the Project increase greenhouse gas emissions over the Baseline Scenario?	Yes	The project will reduce 85,736 tCO ₂ per annum compared to the baseline scenario through replacing fossil fuel-based electricity by renewable energy-based electricity which has nil emissions.	
>>			
Principle 7.2 Energy Supply			
Will the Project use energy from a local grid or power supply (i.e., not connected to a national or regional grid) or fuel resource (such as wood, biomass) that provides for other local users?	Yes	The proposed project generates renewable energy and supplies (except plant's auxiliary consumption) to national grid. The auxiliary consumption is measured by deduction of power supplied to the grid from the total power generated by the plant. The auxiliary consumption is measured by deduction of power supplied to the grid from the total power generated by the plant. However, it's to be noted that the auxiliary power consumed by the plant is generated as renewable energy and thereby no emission involved. Hence, it's not required to be monitored.	Not required
>>			
Principle 8.1 Impact on Natural Water Patterns/Flows			
Will the Project affect the natural or pre-existing pattern of watercourses, ground-	No	As per the section 2 of the EIA, the site is located on the hilly area with no	Not required

water and/or the watershed(s) such as high seasonal flow variability, flooding potential, lack of aquatic connectivity or water scarcity?		water course running through, it is unlikely that the area is flooding or flash flood. Also, the site area is close to sea and land contains small mountains and therefore it wasn't contaminated the groundwater.	
>>			
Principle 8.2 Erosion and/or Water Body Instability			
a. Could the Project directly or indirectly cause additional erosion and/or water body instability or disrupt the natural pattern of erosion? b. Is the Project's area of influence susceptible to excessive erosion and/or water body instability?	Yes	a. As per EIA, during the construction period and heavy rainfall, topsoil erosion may occur due to the excavation activities and run-off respectively. The EPC contractor will develop a Run-off Management Plan for the construction phase, which includes provisions for temporary settlement ponds/basin for surface run-off, especially those during the early stage of a heavy rain. As per 4.1.1.2 of the EIA, the rainwater led into the common stormwater drainage ditches by manholes with screens to retain the large solid wastes and sludge. To make sure the uninterrupted flow of rainwater, the EPC contractor periodically dredged the manholes to avoid blockage during drainage. Therefore, the proposed project doesn't cause	Not required
>>			

		any additional erosion and/or water body instability or disrupt the natural pattern of erosion. b. As per EIA, proper mitigation measures in place to avoid excessive erosion and/or water body instability.	
Principle 9.1 Landscape Modification and Soil			
Does the Project involve the use of land and soil for production of crops or other products?	No	No. Prior to project implementation, the project area was a deserted land contains nearly 70% of hilly and low and medium mountains and remaining 20-30% coastal sands as per the section 2 of the EIA.	Not required
>>			
Principle 9.2 Vulnerability to Natural Disaster			
Will the Project be susceptible to or lead to increased vulnerability to wind, earthquakes, subsidence, landslides, erosion, flooding, drought or other extreme climatic conditions?	No	The purpose of the project activity is to generate clean energy from wind. Hence, it's not susceptible to increased vulnerability to wind, earthquakes, subsidence, landslides, erosion, flooding, drought or other extreme conditions.	Not required
>>			
Principle 9.3 Genetic Resources			
Could the Project be negatively impacted by or involve genetically modified	No	The Project's purpose is to supply clean energy from the	Not required

organisms or GMOs (e.g., contamination, collection and/or harvesting, commercial development, or take place in facilities or farms that include GMOs in their processes and production)?		wind power plant to the national grid. Hence, the proposed project doesn't negatively impact by GMOs	
>>			
Principle 9.4 Release of pollutants			
Could the Project potentially result in the release of pollutants to the environment?	Yes	The project takes a precautionary approach regarding environmental challenges and is not complicit in practices contrary to the precautionary principle. The purpose of the "Law on Environmental Protection" ²³ is to protect the environment with principles of sustainable development and environment. Necessary mitigation measures taken by project owner for environmental safety as follow:	For noise: Inspection of the noise level of works should be carried out at all times; and Reporting of any excessive levels of noise and the measures taken to minimize the impact and prevent from occurring again.
>>			
		For noise during construction phase:	For flickering: Monitoring and reporting of any

²³ <https://thuvienphapluat.vn/van-ban/Tai-nguyen-Moi-truong/Luat-bao-ve-moi-truong-2014-238636.aspx>

		• Undertake noisy activities, including concrete batching and transport to the Site, and installation of the turbines, during daylight hours only. • If noise levels were found to be excessive, construction activities should be stopped until adequate control measures are implemented. • Apply adequate general noise suppressing measures. This could include the use of well-maintained mufflers and noise suppressants for high noise generating equipment and machinery, developing a regular maintenance schedule of all vehicles, machinery, and equipment for early detection of issues to avoid unnecessary elevated noise level, etc. • Ensure that for activities associated with high noise levels, workers are equipped with proper Personal Protective Equipment (e.g., earplugs or earmuffs) • Implement the Grievance Mechanism, including logging of grievances of local people on nuisances caused by excessive noise levels at the local houses caused by the Project to ensure any grievance raised by the local people are properly addressed. For noise during operational phase: • differences in noise levels of less than approximately 2 dBA are generally imperceptible in practice, an increase of 2 dB is hardly perceivable. • differences in noise levels of around 5 dBA are considered to be significant.	complains/grievances on shadow flicker effect during operational phase so as appropriate mitigation measures can be assessed and provided to the sensitive receptors.
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		• differences in noise levels of around 10 dBA are generally perceived to be a doubling (or halving) of the perceived loudness of the noise. An increase of 10 dB is perceived as twice as loud. Therefore, an increase of 20 dB is four times as loud. Mitigation measures for shadow flickering: • The likelihood of direct line of sight to the location of proposed turbine locations will be assessed visually and the potential for using screening such as higher fencing and planting trees will be explored at problematic houses. • The use of curtains or blinds will also be explored. Dust Emission During Construction: The project owner had implemented necessary mitigation measure that is spraying water to suppress the dust on the site as well as the roads periodically. The nearest settlement is 500 meter away from the turbine.	
Principle 9.5 Hazardous and Non-hazardous Waste			
Will the Project involve the manufacture, trade, release, and/ or use of hazardous	Yes	The project doesn't involve the manufacture, trade, release, and/ or	• Inspection of hazardous waste

and non-hazardous chemicals and/or materials?		use of hazardous and non-hazardous chemicals and/or materials. Following mitigation measures has been applied for hazardous waste:	management practices onsite.
>>		• Install separate bins for hazardous waste and ensure they are contained (i.e., covered and not leaked). • Clearly label these bins and indicate what is hazardous waste. • Train staff regarding hazardous waste types and management. • Contract a licensed contractor for the collection, transport, treatment and disposal of hazardous waste. • Follow the requirements for management and storage of hazardous waste as per Circular 36/2015/TT-BTNMT which requires that hazardous waste is stored in a roofed and bunded area with impermeable floor. • Prohibit illegal disposal of hazardous waste to the land. • Ensure that containers are emptied and collected by the contractor at appropriate intervals to prevent overflowing. • Possibly contaminated water (e.g., runoff from paved areas with potential oil spill and leak) must be drained into appropriate facilities (such as temporary sumps and pits). Contaminated drainage must be decanted, and the oily water layer will be skimmed and disposed of as hazardous waste. • Maintain records and manifests that indicate volume of hazardous waste generated onsite, collected treated and	• Review of records and manifests for volume of hazardous waste generated to ensure consistency; and • Regular environmental reporting on implementation of the hazardous waste management practices onsite, including reporting of the hazardous waste management to local DoNRE during the construction (within one month from completion of construction).

		disposed by contractor. The numbers within the records are to be consistent to ensure no illegal discharge at the site or other areas.	
Principle 9.6 Pesticides & Fertilisers			
Will the Project involve the application of pesticides and/or fertilisers?	No	Not applicable for wind power plants.	Not required
>>			
Principle 9.7 Harvesting of Forests			
Will the Project involve the harvesting of forests?	No	No. the project area was an agriculture land where there were no forests.	Not required
>>			
Principle 9.8 Food			
Does the Project modify the quantity or nutritional quality of food available such as through crop regime alteration or export or economic incentives?	No	The project does not modify the quantity or nutritional quality of food available	Not required
>>			
Principle 9.9 Animal husbandry			
Will the Project involve animal husbandry?	No	Not applicable for wind project.	Not required
>>			
Principle 9.10 High Conservation Value Areas and Critical Habitats			

Does the Project physically affect or alter largely intact or High Conservation Value (HCV) ecosystems, critical habitats, landscapes, key biodiversity areas or sites identified?	Yes	Risks of collision for birds and bat when flying into, the rotor sweeping zone. However, this impact significance is expected to be minimal due to low occurrence of avi-fauna at the Site and there is no species of conservation concern. The following mitigation measures adopted: • All tower structures are to be free of holes that can be used for nesting. Roosting habitats (wires and ledges) are to be kept to a minimum. • Shut down-on-demand shall be enabled for all wind turbines. • Contrasting colours on wind turbines in order to make turning blades visible to avifauna • Turbine cut-in speed is to be made slower in order to increase the opportunity for avifauna avoidance during start up.	A carcass monitoring program is to be conducted on a weekly basis at the base of all turbines. A review of the data collected from carcasses monitoring is to be undertaken every 6 months for the first 2 years. Wind farm operations may be altered based on the lifecycle characteristics of any species identified that are susceptible to strike. Assessment of data is to occur yearly from the 3rd year of operation.
Principle 9.11 Endangered Species			
a. Are there any endangered species identified as potentially being present within the Project boundary (including those that may route through the area)? b. Does the Project potentially impact other areas where endangered species	No	a. As per EIA, the site is considered of low ecological significance due to its natural setting and does not impact endangered or rare species.	Not required

may be present through transboundary affects?		b. No. The project does not impact other areas where endangered species may be present.	
>>			

APPENDIX 2- CONTACT INFORMATION OF PROJECT PARTICIPANTS

Organization name	Thuan Nhlen Phong 1 Wind Power Company Limited
Registration number with relevant authority	3401132231
Street/P.O. Box	Hong Chinh Hamlet
Building	Hoa Thang Commune
City	Bac Binh District
State/Region	Binh Thuan Province
Postcode	N/A
Country	Vietnam
Telephone	0933495696
E-mail	tin.h.n@vneco.com.vn
Website	N/A
Contact person	NGUYEN TINH
Title	Director
Salutation	Mr.
Last name	NGUYEN
Middle name	N/A
First name	TINH
Department	N/A
Mobile	0933495696
Direct tel.	0933495696
Personal e-mail	tin.h.n@vneco.com.vn

APPENDIX 3- LUF ADDITIONAL INFORMATION

Not Applicable

APPENDIX 4-SUMMARY OF APPROVED DESIGN CHANGES

Not applicable.

Revision History

Version	Date	Remarks
1.2	14 October 2020	Hyperlinked section summary to enable quick access to key sections Improved clarity on Key Project Information Inclusion criteria table added Gender sensitive requirements added Prior consideration (1 yr rule) and Ongoing Financial Need added Safeguard Principles Assessment as annex and a new section to include applicable safeguards for clarity Improved Clarity on SDG contribution/SDG Impact term used throughout Clarity on Stakeholder Consultation information required Provision of an accompanying Guide to help the user understand detailed rules and requirements
1.1	24 August 2017	Updated to include section A.8 on 'gender sensitive' requirements
1.0	10 July 2017	Initial adoption